CardioCog: An Explainable AI Framework for Cognitive Health Assessment and Early Prediction of Cognitive Decline Using Cardiovascular, Genetic, and Clinical Factors

FINAL REVIEW – B.TECH PROJECT

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Research progression

Zeroth review: multimodal framework → first review: 3D amyloid-PET prototype (198 people, 17 events) → full NACC clinical cohort (19,470 people, 2,818 events) → two decline outcomes, five models, fusion network → PET image pipeline, prototype, 17 verification checks.

Milestone	**Evidence and outcome**
First-review prototype	3D CNN + cross-attention on amyloid PET: AUROC 0.786 in one run, 0.652 on re-run. Too few events to conclude.
Leakage-free cohort	First visit, not demented, UDS v1–3; person-level split with 9 whole centres held out.
Dementia in 3 years	XGBoost AUROC 0.938 (test), 0.946 (held-out centres); 0.84 within MCI.
Cognitive decline	CDR-SB rise ≥ 1: AUROC 0.858 / 0.885; CDR-SB change regression helps MCI.
Independent checks	17 of 17 checks pass; two future-information errors found and fixed.

Problem

Clinics already record cognition, vascular history, genetics and sometimes PET for older adults without dementia, but rarely combine them into one *explainable* estimate of near-term decline. Which of these sources actually adds information, and can the estimate be trusted?

For a person without dementia at visit t_0, estimate over the next three years:

$$\hat{p}_{\text{dem}} = P(\text{dementia} \mid x_{t_0}), \quad \hat{p}_{\text{dec}} = P(\Delta\text{CDR-SB} \geq 1 \mid x_{t_0}), \quad \hat{\Delta} = E(\Delta\text{CDR-SB} \mid x_{t_0})$$

1. How accurately can both outcomes be predicted, for normal and MCI groups separately?
2. Does cardiovascular history add to demographics, genetics and current cognition?
3. Do amyloid and tau PET add to the clinical model?
4. Do deep models, including a modality-encoder fusion network, beat classical ones?
5. Do earlier visits help? 6. Can one model cope with missing categories? 7. Can each estimate be explained with evidence?

Work	Core method	Relevance / limitation for CardioCog
Livingston et al. (2020)	Lancet Commission, 12 modifiable risk factors	Vascular risk matters over the life course; mostly midlife exposure
Alaka et al. (2025)	CAIDE + heart rate, random forest, NACC	Same database; AUC 0.80–0.91, no within-stage results
Stern & Linial (2026)	CatBoost on UK Biobank codes + APOE	AUROC 0.773 without cognitive inputs
Aschwanden et al. (2020)	Random survival forests, 52 predictors	Cardiovascular factors less important than expected
Yi et al. (2023)	XGBoost + SHAP for diagnosis	CDR-SB most important; diagnosis, not prediction
Gorishniy et al. (2021)	FT-Transformer for tabular data	No deep model consistently beats boosted trees
Lewis et al. (2020)	Retrieval-augmented generation	Grounds generated text in a fixed library

Research gap

Few studies measure, on one large leakage-controlled cohort, **how much each source adds once current cognition is known**, with results **within** normal and MCI groups.

Why the design changed after the first review

First-review prototype

3D CNN on raw amyloid PET + 34 clinical features, spatial cross-attention, 1-year CDR-SB rise ≥ 0.5.

Run	Mean AUROC (SD)
Reported at review	0.786 (0.134)
Re-run, same notebook	0.652 (0.153)

Diagnosis

Only 198 people and **17 events**: a handful of participants decides the score. The pipeline also mixed PET time frames, had no registration, read one non-existent column and fitted SHAP on all data.

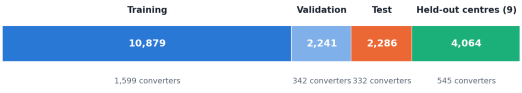
Decision

Move to the full clinical cohort with three-year outcomes and thousands of events; treat PET images as a separate, validated pipeline.

Step	Participants
All participants	57,038
First visit, in person, UDS v1–3	55,461
not demented at that visit	38,107
outcome known at 3 years	19,470
converted to dementia	2,818 (14.5%)

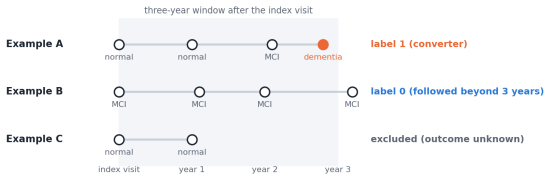
Three groups, very different risk

Normal 12,377 (1.6% convert)
MCI 5,993 (41.5%)
Impaired, not MCI 1,100 (12.3%)



- Person-level split; **9 of 46 centres held out** entirely.
- Train 70% / validation 15% / test 15%, seed 2026.
- Every normaliser and imputer fitted on training data only.

Outcomes and censoring



$$y = \begin{cases} 1 & T \le 3 \\ 0 & T > 3,\ C \ge 3 \\ \text{unknown} & T > 3,\ C < 3 \end{cases}$$

T: years to dementia diagnosis, C: years to last visit.

Secondary outcomes

CDR-SB rise ≥ 1 within 3 years (20,355 people, 5,035 events) and the 3-year change Δ in CDR-SB.

Robustness

A Cox model that keeps censored participants gives the same conclusions.

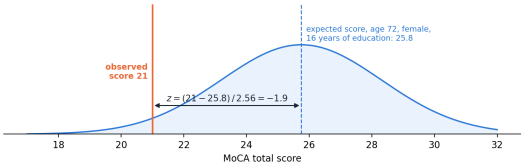
Category	Features
Demographics	8
Cardiovascular and medical	25
Cognitive tests (z-scores)	9
Clinical stage (CDR-SB, diagnosis)	3
Genetics (APOE $\varepsilon 4$, family history)	3
PET (second-stage model only)	4

Harmonising tests across form versions

$$\hat{s} = \beta_0 + \beta_1\,\text{age} + \beta_2\,\text{female} + \beta_3\,\text{educ}$$

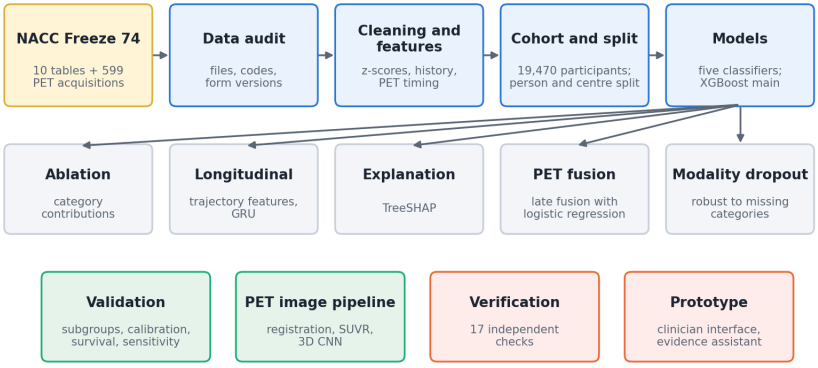
$$z = (s - \hat{s})/\sigma_{\text{res}}$$

Fitted on cognitively normal training participants.



MoCA example: expected 25.8 for a 72-year-old woman with 16 years of education, so a score of 21 gives $z = -1.9$. Healthy people not used for fitting score within 0.1 of zero in every UDS version.

06 | Methodology: the CardioCog Pipeline



Rule for every step

Only information available at the index visit; anything learned from data is fitted on the training set; results are reported on the test set and on centres never seen in training.

Models compared on the same 48 features

Model	Key settings
Logistic regression	L_2, C from 5 values
Random forest	500 trees, $\sqrt{p}$ features
XGBoost	$\eta = 0.03$, depth 3–6, $\lambda = 2$, early stop
MLP	128–64 units, dropout 0.2
FT-Transformer	64-d tokens, 3 layers, 8 heads

Three seeds averaged; XGBoost chosen as the main model (handles missing values, exact SHAP).

XGBoost objective

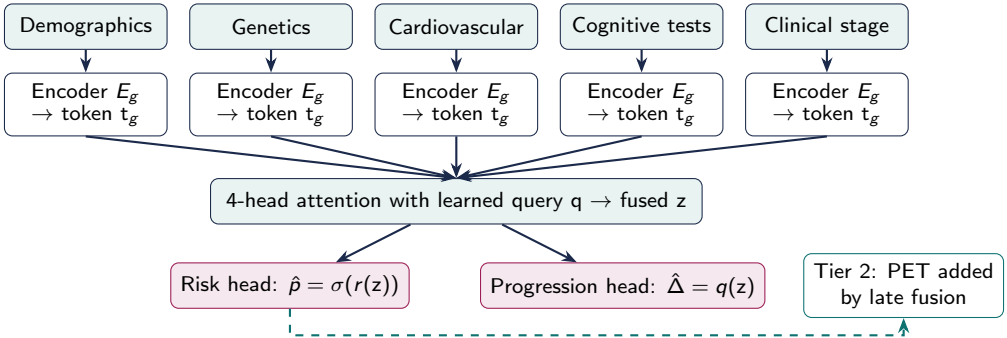
$$f_M(\mathrm{x}) = \sum_{m=1}^{M} \eta\, g_m(\mathrm{x}), \qquad w_j^* = -\frac{\sum_{i \in I_j} G_i}{\sum_{i \in I_j} H_i + \lambda}$$

Loss and evaluation

$$\mathcal{L}_{\mathrm{BCE}} = -\tfrac{1}{n} \textstyle\sum_i \left[y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \right]$$

AUROC, AUPRC, Brier, calibration slope, net benefit; 1,000 paired bootstrap resamples.

Modality-encoder fusion network (proposed architecture)



A missing modality gets a learned token m_g; each token gets a type embedding τ_g; whole modalities are hidden with probability 0.35 in training.

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + 0.5 \cdot \text{Huber}(\Delta - \hat{\Delta}), \qquad \alpha_g = \text{softmax}_g\big((W_Q q)^\top W_K t_g / \sqrt{8}\big)$$

07 | Results: Dementia Within Three Years

	Test set (2,286 / 332 events)			Held-out centres (4,064 / 545)	
Model	AUROC (95% CI)	AUPRC	Brier	AUROC	Brier
Logistic regression	0.936 (0.924–0.949)	0.721	0.066	0.945	0.060
Random forest	0.932 (0.919–0.944)	0.704	0.070	0.940	0.064
XGBoost	**0.938** (0.925–0.948)	0.713	0.066	**0.946**	0.059
MLP	0.938 (0.926–0.949)	0.721	0.066	0.946	0.059
FT-Transformer	0.939 (0.927–0.949)	0.720	0.066	0.945	0.061

What the result does establish

All five models discriminate well and are well calibrated (constant-prediction Brier 0.124), at new centres as well as on the test set.

What it does not establish

Intervals overlap almost completely: deep models give **no gain** over logistic regression or boosted trees.

Is 0.94 too good? Results within diagnostic groups

Group (test / held-out)	Test	Held-out
All not demented	0.937	0.946
Cognitively normal (24 / 37 ev.)	0.884	0.876
MCI (289 / 496 ev.)	0.835	0.846

Interpretation

Diagnosis alone gives AUROC 0.84, because MCI converts 25 times as often as normal cognition. Within MCI the model still ranks risk at **0.84**.

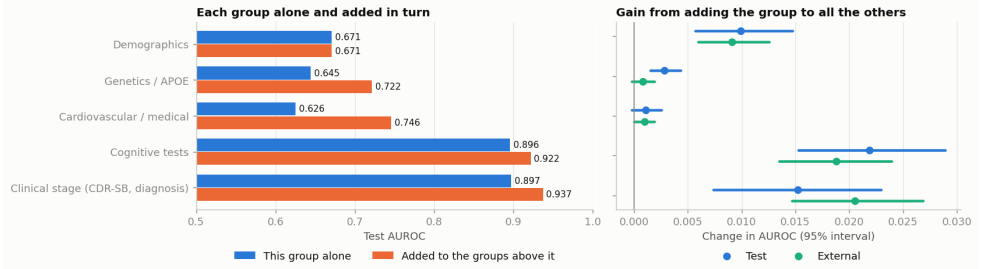
Critical check

Removing *all* current-cognition information drops AUROC from 0.937 to **0.746**.

Meaning

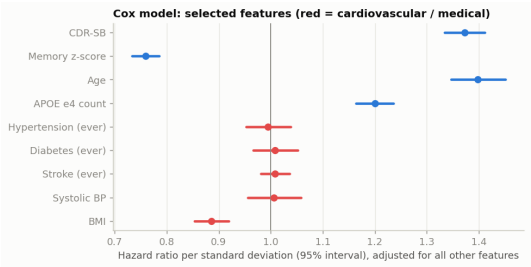
The model estimates short-term progression *given* the current clinical stage. It is not leakage: inputs are from the index visit, the label from later visits.

What drives the prediction? Category ablation



Category	Alone	Gain added last (test)	Gain (held-out)
Cognitive tests	0.896	+0.022	+0.019
Clinical stage	0.897	+0.015	+0.021
Demographics	0.671	+0.010	+0.009
Genetics (APOE)	0.645	+0.003	+0.001
Cardiovascular / medical	0.626	+0.001	+0.001

The cardiovascular question (RQ2)

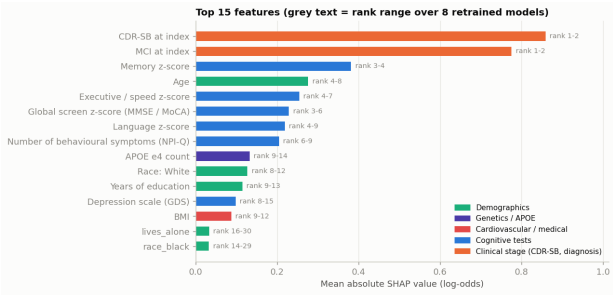


- Alone: AUROC **0.626**; before cognition: +0.024 (test), +0.004 (held-out).
- After cognition: +0.001 for dementia, +0.0004 for CDR-SB decline.
- Cox model: hypertension HR 0.995, diabetes and stroke ≈ 1.

Interpretation

Limited *incremental* value in this cohort, not evidence that vascular health is unimportant: median age 71, midlife exposure unobserved, and past damage already shows in the cognitive scores.

Explainability: Shapley values

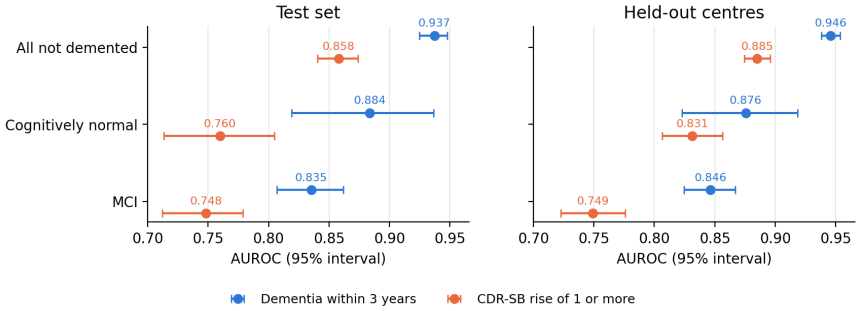


$$f(x) = \phi_0 + \sum_j \phi_j$$

Category	SHAP share
Clinical stage	40.8%
Cognitive tests	35.0%
Demographics	15.5%
Cardiovascular	5.1%
Genetics	3.5%

Stable over 8 refits (Spearman 0.89–0.97). Low BMI raises risk: weight loss before diagnosis, not causal.

Cognitive decline as the outcome



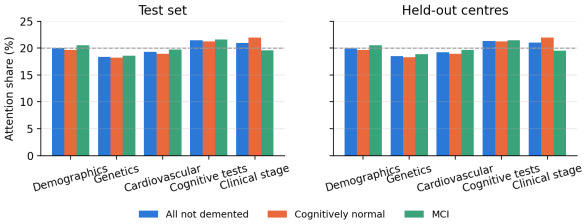
CDR-SB rise ≥ 1 (20,355 / 5,035 events)

AUROC 0.858 test, 0.885 held-out; normal 0.76–0.83 with 129 and 215 events, so more reliable than the dementia estimate for normal cognition.

Predicted 3-year CDR-SB change

MAE 0.69 $\rightarrow$ 0.60 (test), 0.82 $\rightarrow$ 0.72 (held-out); MCI 1.90 $\rightarrow$ 1.59; normal 0.17 = 0.17, no gain.

Fusion network: works, but no gain and uninformative attention



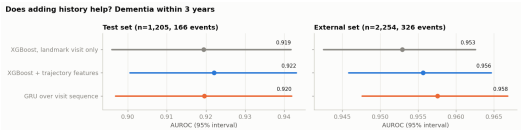
AUROC	Fusion net	XGBoost
Test	0.934	0.937
Held-out	0.943	0.946
MCI (test)	0.831	0.835

Progression head MAE 0.72 $\rightarrow$ 0.65. Without cognition: AUROC 0.72, Brier 0.114.

Finding

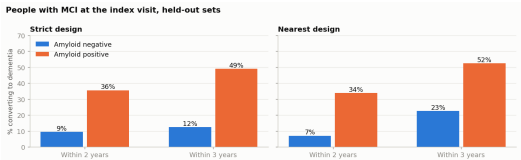
Attention is 18–22% for every modality (uniform = 20%), so it does not explain predictions. SHAP and ablation are used instead.

Earlier visits and PET measures (RQ3, RQ5)



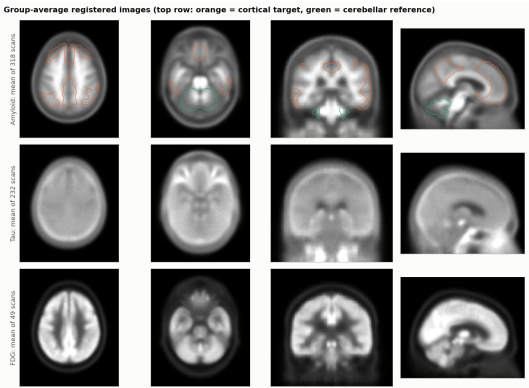
Visit history

Landmark visits with ≥ 2 earlier visits: trajectory features $+0.003$, GRU $+0.000$ to $+0.005$ AUROC.



Amyloid and tau PET

MCI: 34% of amyloid-positive vs 7% of negative convert within 2 years. Late fusion: 0.964 vs 0.944 on 243 people (33 events), gain interval -0.009 to $+0.066$ includes zero.



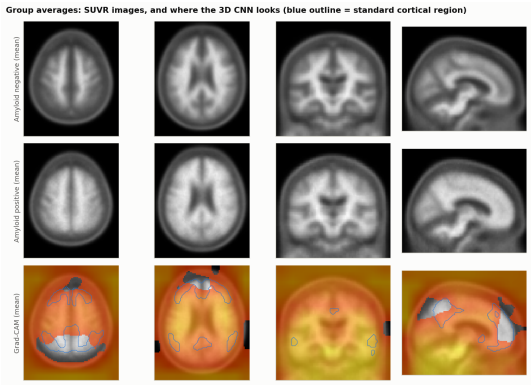
Group averages only (318 amyloid, 232 tau, 49 FDG scans).

1. **Convert:** 599 dynamic DICOM series $\rightarrow$ static volume, $V = \sum_{k \in \mathcal{W}} d_k V_k / \sum_{k \in \mathcal{W}} d_k$.
2. **Register:** rigid to MNI152, then 12-parameter affine to a study PET template, Mattes mutual information.
3. **Quantify:** Centiloid masks, $\mathrm{SUVR} = \bar{V}_{\mathrm{cortex}} / \bar{V}_{\mathrm{cerebellum}}$.

Validation against the NACC PET core

Pearson $r = \mathbf{0.977}$ over 318 scans (0.96–0.99 per tracer); mean difference -0.043.

3D CNN for amyloid status and Grad-CAM



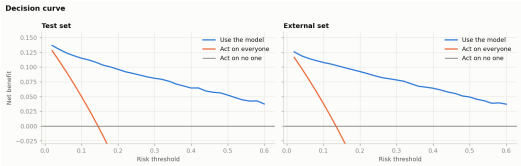
Method (292 scans, 84 positive)	AUROC
Cortical SUVR (one number)	**0.982**
Regional logistic model	0.971
3D CNN (494,917 weights)	0.960

Interpretation

The pipeline supports a working image model, but the label is itself derived from cortical SUVR, so one number is hard to beat.

Limitation

Grad-CAM was diffuse (1.7% of top voxels in the cortical region vs 3.8% by chance), and only 5 imaged people converted, so images were not used to predict dementia.



- Test: predicted 14.5% = observed 14.5%, slope 1.01.
- Held-out: 15.0% vs 13.4%, slope 1.10.
- At 20% risk: sensitivity 83%, specificity 88%, NPV 97%.
- Net benefit above treat-all and treat-none from 2% to 60%.

Brier score	Standard	Dropout	Specialist
Everything	0.066	0.067	0.066
No clinical stage	0.185	0.076	0.074
No cognition at all	0.441	0.115	0.111
Demographics only	0.444	0.120	0.118

Modality dropout ($p = 0.35$)

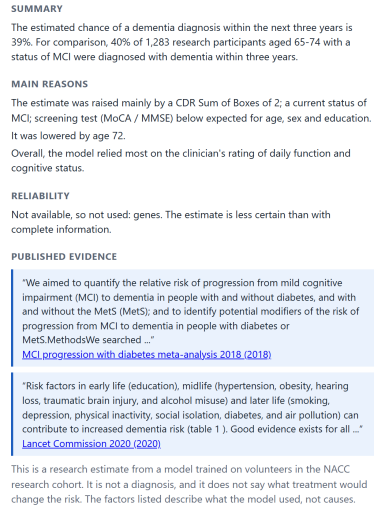
One model stays close to a specialist retrained for each missing pattern; a standard model becomes worse than the base rate.



Example: 72-year-old woman, 16 years of education, MCI, CDR-SB 2, MoCA 21.

- FastAPI + React; form generated from the 48 trained features.
- Risk shown next to the observed rate of similar people, with SHAP by category.
- Plausibility checks flag contradictory entries.
- Plain-language report with quoted evidence.
- 27 of 28 behaviour checks pass; MCI under 65 over-predicted (35% vs 27%).

Plain-language report for the same patient



SUMMARY

The estimated chance of a dementia diagnosis within the next three years is 39%. For comparison, 40% of 1,283 research participants aged 65-74 with a status of MCI were diagnosed with dementia within three years.

MAIN REASONS

The estimate was raised mainly by a CDR Sum of Boxes of 2; a current status of MCI; screening test (MoCA / MMSE) below expected for age, sex and education.

It was lowered by age 72.

Overall, the model relied most on the clinician's rating of daily function and cognitive status.

RELIABILITY

Not available, so not used: genes. The estimate is less certain than with complete information.

PUBLISHED EVIDENCE

"We aimed to quantify the relative risk of progression from mild cognitive impairment (MCI) to dementia in people with and without diabetes, and with and without the MetS (MetS); and to identify potential modifiers of the risk of progression from MCI to dementia in people with diabetes or MetS.MethodsWe searched ..."
MCI progression with diabetes meta-analysis 2018 (2018)

"Risk factors in early life (education), midlife (hypertension, obesity, hearing loss, traumatic brain injury, and alcohol misuse) and later life (smoking, depression, physical inactivity, social isolation, diabetes, and air pollution) can contribute to increased dementia risk (table 1). Good evidence exists for all ..."
Lancet Commission 2020 (2020)

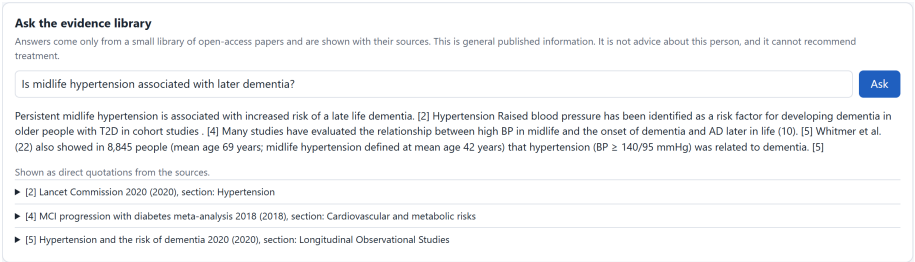
This is a research estimate from a model trained on volunteers in the NACC research cohort. It is not a diagnosis, and it does not say what treatment would change the risk. The factors listed describe what the model used, not causes.

1. **Summary:** risk next to the observed rate of similar people (39% vs 40%).
2. **Main reasons:** largest SHAP contributions in plain words.
3. **Reliability:** missing categories and plausibility warnings.
4. **Published evidence:** quoted passages with their source.
5. **Limits:** fixed statement that it is not a diagnosis.

Design choice

Templates are filled with the person's own numbers and evidence is *quoted*, so the report cannot state anything the model or the literature did not provide.

Evidence assistant (retrieval-augmented generation)



$$s_{\mathrm{RRF}}(d) = \frac{1}{60 + \rho_{\mathsf{sem}}(d)} + \frac{1}{60 + \rho_{\mathrm{BM25}}(d)}$$

MiniLM embeddings + BM25 ($k_1 = 1.5$, $b = 0.75$), merged by rank fusion. Below similarity 0.35 the assistant says the library has no evidence.

Test (11 papers, 447 passages)	Result
Correct source first: hybrid	32 / 32
embeddings only / BM25 only	28 / 31
Personal-advice questions flagged	8 / 8
Out-of-scope questions declined	6 / 6

11 | Independent Verification

Check	Result
No person or held-out centre in more than one partition	0 shared
Features unchanged when all later visits are deleted	Identical for 1,500 people $\times$ 67 features
No PET value newer than the visit that uses it	0 of 1,481 amyloid, 713 tau
Labels recomputed independently	0 disagreements in 11,403 visits
Shuffled labels	AUROC 0.508 (chance)
Retrain with a new seed	0.937 / 0.946 (reported 0.938 / 0.946)

Errors the checks found and we fixed

Education and race filled from a *later* visit (0.1% of people); PET scans matched to visits weeks *before* the scan. After fixing, every model was retrained: results moved only in the third decimal. **17 of 17 checks pass.**

Synthesis: what each research question found

Research question	Answer and evidence
RQ1 Prediction accuracy	Dementia 0.94 (0.84 in MCI); decline 0.86 (0.75 in MCI); change regression helps MCI only
RQ2 Cardiovascular history	+0.001 AUROC once cognition is known; Cox agrees
RQ3 Amyloid and tau PET	Strong association, no demonstrable gain with current follow-up
RQ4 Deep models / fusion net	No gain over logistic regression or XGBoost; attention uniform
RQ5 Earlier visits	+0.000 to +0.005 AUROC
RQ6 Missing categories	Modality dropout keeps calibration close to specialist models
RQ7 Explanation and evidence	SHAP, plausibility checks, report with quoted published evidence

Final takeaway

How a person is functioning today carries most of the three-year signal; decades of vascular and genetic history leave their trace in that measurement rather than adding to it.

Limitations, Future Work and Final Takeaway

- Held-out centres are site-level validation within NACC; external and temporal validation on another cohort is the next step.
- NACC is a research cohort, mostly White and highly educated; about half of eligible people lacked a 3-year outcome (Cox check agrees).
- Only 24 and 37 cognitively normal converters in test and held-out sets; a dedicated 5- to 10-year model for normal cognition is needed.
- Richer vascular exposure (repeated blood pressure, treatment, small-vessel imaging) may capture what yes/no history misses.
- PET registration without MRI, tau not quantified; more follow-up will allow a powered PET test.
- Evidence library of 11 papers and team-written questions; a curated library and stronger local model would help.

Contribution

A leakage-free, verified pipeline that measures what each source of information adds, reports results within diagnostic groups, and delivers explainable, evidence-backed estimates.

Thank you

Questions are welcome.

Data: National Alzheimer's Coordinating Center (NACC), Freeze 74, used under the NACC data use agreement.
All results are aggregates.